

Perfect Subset Privacy in Polynomial Computation

Zirui (Ken) Deng^{*}, Vinayak Ramkumar[†], and Netanel Raviv^{*}

^{*}Department of Computer Science and Engineering, Washington University in St. Louis, St. Louis, MO, USA

[†]Department of Electrical Engineering–Systems, Tel Aviv University, Tel Aviv, Israel
d.ken@wustl.edu, vinram93@gmail.com, netanel.raviv@wustl.edu

Abstract—Delegating large-scale computations to service providers is a common practice which raises privacy concerns. This paper studies information-theoretic privacy-preserving delegation of data to a service provider, who may further delegate the computation to auxiliary worker nodes, in order to compute a polynomial over that data at a later point in time. We study techniques which are compatible with robust management of distributed computation systems, an area known as *coded computing*. Privacy in coded computing, however, has traditionally addressed the problem of colluding workers, and assumed that the server that administrates the computation is trusted. This viewpoint of privacy does not accurately reflect real-world privacy concerns, since normally, the service provider as a whole (i.e., the administrator and the worker nodes) form one cohesive entity which itself poses a privacy risk. This paper aims to shift the focus of privacy in coded computing to safeguarding the privacy of the user against the service provider as a whole, instead of merely against colluding workers inside the service provider. To this end, we leverage the recently defined notion of *perfect subset privacy*, which guarantees zero information leakage from all subsets of the data up to a certain size. Using known techniques from Reed-Muller decoding, we provide a scheme which enables polynomial computation with perfect subset privacy in straggler-free systems. Furthermore, by studying *information super-sets* in Reed-Muller codes, which may be of independent interest, we extend the previous scheme to tolerate straggling worker nodes inside the service provider.

Index Terms—Perfect subset privacy, distributed systems, coded computing, Reed-Muller codes.

I. INTRODUCTION

As significant computational power is no longer within the grasp of individual users, relying on third-party services for data storage and computation has become a necessity. While this paradigm has commendably allowed smaller, non-technological entities to harness capabilities beyond their means, it inevitably poses risks related to the privacy of these entities.

In recent years, there has been considerable interest in applications of error-correcting codes in distributed computations, commonly known as *coded computing*, in order to tackle straggling (i.e., slow) workers, adversarial workers, and privacy issues in such systems [14, 21, 22]. Privacy guarantees, however, have typically been given in terms of some level of restricted collusion among workers, while the system administrator is assumed to be trusted.

Parts of this work were done when V. Ramkumar was a postdoctoral researcher in the Department of Computer Science and Engineering in Washington University in St. Louis.

This approach overlooks the common setting where a data owner relies on a single *untrusted* service provider in order to store and compute over their data, and the service provider itself inherently introduces a privacy risk. Whether the storage or computation is distributed internally or not, the service provider has access to the entirety of the data, rendering the assumption of limited collusion among workers largely irrelevant. This scenario is widely prevalent, as many institutions delegate their data in its entirety to service providers such as AWS (Amazon Web Services) or Google Cloud. In this work, we seek to extend the scope of coded computing and develop new techniques to protect privacy against the service provider as a whole, while enabling the service provider’s internal system administrator (abbrev. admin) to distribute the computation internally.

In our setup, we assume the user has some data $\mathbf{x}$, which is internally partitioned to n parts $x_1, \dots, x_n$ of some contextual meaning (e.g., features). The service provider consists of a system admin and workers, among which collusion is not restricted. The user cannot communicate with the workers directly, but rather through the system admin. The user encodes $\mathbf{x}$ using some random key vector $\mathbf{k}$ and then sends the encoded (privatized) version of the data $\tilde{\mathbf{x}}$ to the admin, who in turn distributes $\tilde{\mathbf{x}}$ to multiple workers. Due to storage limitations, the user wishes to discard $\mathbf{x}$ and instead only keep the key $\mathbf{k}$ as side information. At a later point in time, the user is interested in computing some polynomial f over $\mathbf{x}$, and sends f to the admin. The goal of the admin is to reply to the user with some information $\mathcal{A}$, computed with the aid of the workers, so that the user can extract the desired computation results, potentially using $\mathbf{k}$. The purpose of this work is to shift the “privacy barrier” in coded computing: instead of protecting the privacy of the admin against colluding workers, we wish to protect the privacy of the user against the service provider as a whole. We call the above the *single-provider setup*, illustrated in Figure 1.

We aim to provide information-theoretic privacy guarantees. However, bounding the mutual information between the actual and privatized data provides no understanding about which parts of the data are revealed, and to what extent. To remedy that, we leverage the recently developed notion of *perfect subset privacy* [18], which guarantees zero leakage from *all* subsets of the data up to a certain size r . This notion provides a uniform privacy metric that is easy to understand, and r serves as a tunable privacy parameter that trades better privacy for higher storage and computation requirements, and vice versa.

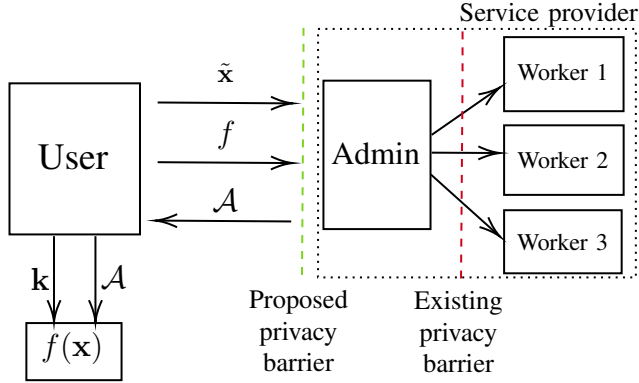


Fig. 1: Shifting the privacy barrier in coded computing.

We first generalize the encoding scheme from [18] to apply to finite fields of any size. Next, by utilizing the known notion of *information sets* in Reed-Muller decoding, we devise a scheme which allows the admin to compute f over the encoded data $\tilde{x}$, either locally or in a system without straggling workers; information sets are used specifically to alleviate the download cost of the scheme. Then, in order to adapt the above scheme to tolerate straggling workers, we study a notion that we refer to as *information super-sets* for Reed-Muller codes. This notion may be of independent interest, and can be seen as related to the problem of puncturing Reed-Muller codes efficiently. We proceed by devising good straggler-resilient schemes by constructing information super-sets for binary Reed-Muller codes.

The rest of the paper is structured as follows. The problem setup and computation model are given in Section II. An encoding procedure that achieves perfect subset privacy is provided in Section III. In Section IV we reduce the user's download cost in a straggler-free system. In Section V we devise straggler-resilient protocols using information super-sets.

II. PROBLEM STATEMENT

A. The computation model

Our distributed computation model consists of a storage phase and a computation phase. In what follows, the user holds uniformly distributed data $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}_q^n$ chosen from $X = \text{Unif}(\mathbb{F}_q^n)$ and encodes the data into a privatized version prior to sharing it. Specifically, the user encodes $\mathbf{x}$ to $\tilde{\mathbf{x}} \triangleq \tilde{E}(\mathbf{x}, \mathbf{k})$, where $\tilde{E}$ is some encoding function and $\mathbf{k} \in \mathbb{F}_q^m$, chosen from $K = \text{Unif}(\mathbb{F}_q^m)$, is some random key, with $m \ll n$. Then, the user sends $\tilde{\mathbf{x}}$ to the admin, who further encodes $\tilde{\mathbf{x}}$ to $\hat{\mathbf{x}} \triangleq \hat{E}(\tilde{\mathbf{x}}) = (\hat{x}_1, \dots, \hat{x}_N)$, where $\hat{E}$ is another encoding function. The admin distributes $\hat{\mathbf{x}}$ to N workers, with worker $i \in [N]$ (where $[N] \triangleq \{1, 2, \dots, N\}$) storing $\hat{x}_i$. The user wishes to discard $\mathbf{x}$ and only keep $\mathbf{k}$ as side information, which requires less storage. We call this the *storage phase* of the model.

At a later point, the user is interested in computing $f(\mathbf{x})$, where $f : \mathbb{F}_q^n \rightarrow \mathbb{F}_q$ is a multivariate polynomial of total degree at most d and individual degree at most $q - 1$ (i.e., $d < n(q - 1)$). The polynomial f is not known during the storage phase. The user sends f to the admin, who then sends it to the N workers. For $i \in [N]$, worker i performs some computation over $\hat{x}_i$, and returns the results to the admin. The admin aggregates the results into some information $\mathcal{A}$ that is sent back to the user; $\mathcal{A}$ along with side information $\mathbf{k}$ should suffice for decoding $f(\mathbf{x})$. We call this the *computation phase* of the model. Among those N workers, however, S ones may straggle, i.e., not return a response in a timely manner, and the admin must be able to provide the user with sufficient information to decode $f(\mathbf{x})$ even in the absence of the computation results from the stragglers.

Remark 1. For simplicity, we assume the user is interested in computing f on a single $\mathbf{x}$, unlike the common system assumption of multiple $\mathbf{x}$'s [22]. Our setting can be generalized to multiple $\mathbf{x}$'s (e.g., a dataset with multiple datapoints, or a matrix with many columns) by straightforward repetition.

B. Perfect subset privacy

Let $\tilde{X}$ be the random variable which is induced by the data distribution X , the distribution of the key K , and the function $\tilde{E}$, i.e., $\tilde{X} = \tilde{E}(X, K)$. An encoding scheme is called *perfectly private* [18] if the transformed and original data have zero mutual information, i.e., $I(X; \tilde{X}) = 0$. Perfect privacy is a very strong notion, which requires a key as big as the data itself [5]; our scheme would trivialize under such requirements, as no storage saving would be possible. Therefore, we must resort to limited information leakage.

However, guarantees of the form $I(X; \tilde{X}) \leq \epsilon$ are rather vague, since they provide no insight about which part of the data is revealed. To this end, we leverage the notion of *perfect subset privacy*, first developed in [18] by generalizing *perfect sample privacy* of [16]. To define this notion, let $\binom{[n]}{r}$ be the collection of all subsets of $[n]$ of size r . For any $\mathcal{R} \subseteq \binom{[n]}{r}$, we use the notation $X_{\mathcal{R}} \triangleq (X_i)_{i \in \mathcal{R}}$, and later extend this notation to matrices and their columns in a natural way.

Definition 1. [18, Def. 1] For a privacy parameter $r \in [n]$, an encoding procedure is said to satisfy r -subset privacy if $I(X_{\mathcal{R}}; \tilde{X}) = 0$ for all $\mathcal{R} \in \binom{[n]}{r}$.

This notion of r -subset privacy provides uniform protection for all r -subsets of the data with a tunable parameter r . It is easy to show [16, 18] that r -perfect subset privacy implies that $I(X; \tilde{X}) \leq (n - r) \log_2 q$ bits. Therefore, it can be seen as a special case of bounding information leakage, which has additional uniform and clear guarantees regarding the structure of the privacy protection. It is to be noted that perfect subset privacy and the popular notion of *differential privacy* [6] are two different privacy paradigms, and that neither implies the other, as explained in [18]. We also remark that perfect subset privacy is different from collusion resistance [3, 22], which requires $I(X; \tilde{X}_{\mathcal{R}}) = 0$ for all $\mathcal{R} \in \binom{[n]}{r}$. Furthermore, perfect

subset privacy is to be distinguished from the notion of *individual security* [2, 4, 13, 20], which states that $I(\tilde{X}_{\mathcal{R}}; X_{\mathcal{Q}}) = 0$ for all $\mathcal{R} \in \binom{[n]}{r}$ and $\mathcal{Q} \in \binom{[n]}{n-r}$. Neither of these notions is relevant for our problem definition, since in our setting the service provider has access to $\tilde{X}$ in its entirety.

C. Performance metrics

We say that any scheme that realizes the computation model in Section II-A is an (n, q, r, d, S) -scheme, where n is the length of the data $\mathbf{x}$ held by the user, q is the field size, r is the privacy parameter, d is the maximum total degree of f , and S is the maximum number of stragglers in the system. Below are the relevant metrics by which we judge the merit of such schemes.

- **The number of workers**, measured by the parameter N .
- **Download cost**, measured by the number D of $\mathbb{F}_q$ symbols the user needs to download in order to complete the decoding process.
- **Side information**, measured by m , the size of the random key $\mathbf{k}$. As mentioned before, we would like m to be much smaller than n , given the limited storage capability of the user.

We would like to minimize all three quantities, and indeed, Section III deals with minimizing the amount of side information held by the user, Section IV addresses the problem of reducing download cost, and Section V optimizes the number of workers needed for computation.

III. OBTAINING r -SUBSET PRIVACY

In this section, we present an encoding procedure which achieves perfect subset privacy. In [18], Raviv et al. presented a method that obtains r -subset privacy when the finite field is sufficiently large. We first give a brief description of their method and then generalize it to finite fields of any size.

For distinct elements $w_1, \dots, w_n \in \mathbb{F}_q$, let $\tilde{\mathbf{x}} \triangleq \mathbf{x} + \mathbf{k}V$, where $\mathbf{k} \sim \text{Unif}(\mathbb{F}_q^r)$ and $V = V(w_1, \dots, w_n)$ is an $r \times n$ Vandermonde matrix over $w_1, \dots, w_n$. This method has been proven [18, Thm. 1] to achieve r -subset privacy, but it requires $q \geq n$.

We now generalize this method to obtain a procedure that guarantees subset privacy without any field size restriction. Specifically, let

$$\tilde{\mathbf{x}} \triangleq \mathbf{x} + \mathbf{k}G, \quad (1)$$

where $\mathbf{k} \sim \text{Unif}(\mathbb{F}_q^m)$ and $G \in \mathbb{F}_q^{m \times n}$ is a generator matrix of any $[n, m]_q$ linear code $\mathcal{C}$ that satisfies $d_{\min}(\mathcal{C}^\perp) \geq r + 1$, where $d_{\min}(\cdot)$ denotes minimum Hamming distance and $\mathcal{C}^\perp$ is the dual code of $\mathcal{C}$. This implies that for all $\mathcal{R} \in \binom{[n]}{r}$, $G_{\mathcal{R}} \in \mathbb{F}_q^{m \times r}$ has full rank.

Remark 2. Notice that $m \geq r$ according to the Singleton bound, and the equality is achieved with Maximum Distance Separable (MDS) codes. For example, the scheme in [18] achieves equality with Reed-Solomon codes whose generator matrices are Vandermonde matrices.

Remark 3. In our encoding scheme, there is an inherent tradeoff between field size and number of required keys. For example, say $n = 16$ and $r = 3$, which implies that $d_{\min}(\mathcal{C}^\perp) \geq r + 1 = 4$. If $q = 2$, then $m = 5$ keys are needed [10], whereas if we consider the field $GF(8)$, then only four keys are necessary. The minimum number of keys needed is 3, achieved by Reed-Solomon codes when the field size is 16.

The following theorem establishes that this generalized encoding indeed achieves perfect subset privacy. See the full version of the paper [8] for a proof.

Theorem 1. For any $r \in [n]$, the encoding procedure in (1) satisfies r -subset privacy, i.e., $I(X_{\mathcal{R}}; \tilde{X}) = 0$ for all $\mathcal{R} \in \binom{[n]}{r}$.

It follows from (1) that $\tilde{\mathbf{x}} \in \mathbf{x} + \mathcal{C}$, i.e., that $\mathbf{x}$ and $\tilde{\mathbf{x}}$ belong to the same coset of $\mathcal{C}$. Therefore, it is possible to reduce upload cost from n to $n - m$ by communicating to the service provider the identity of the coset of $\mathcal{C}$ containing $\mathbf{x}$, e.g., by computing a syndrome, which in turn identifies a shift vector that defines the coset². Then, the service provider is able to perform computations based on the syndrome and respond to the user accordingly. Further details can be found in the full version [8].

Now that we have demonstrated an encoding scheme which achieves r -subset privacy for any field size q , we turn our attention to devising a compatible computation scheme.

IV. LOWERING DOWNLOAD COST IN STRAGGLER-FREE SYSTEMS

In this section, we first present a trivial scheme which serves as a benchmark. Then, we show that it is possible to reduce download cost through evaluation of a dedicated polynomial at an information set for Reed-Muller codes.

The trivial scheme works as follows. The user sends $\tilde{\mathbf{x}}$ as defined in (1) to the admin. The admin encodes $\tilde{\mathbf{x}}$ to $\{\tilde{\mathbf{x}} - \mathbf{t}G | \mathbf{t} \in \mathbb{F}_q^m\}$ and distributes these to the workers in the storage phase. Later in the computation phase, the admin gets f from the user. The admin sends the answer $\mathcal{A} \triangleq (f(\tilde{X} - \mathbf{t}G) | \mathbf{t} \in \mathbb{F}_q^m)$, and then the user can retrieve the element that corresponds to the correct key $\mathbf{k} \in \mathbb{F}_q^m$ and get the desired result $f(\mathbf{x}) = f(\tilde{\mathbf{x}} - \mathbf{k}G)$. The download cost D of the trivial scheme equals q^m . This scheme is termed “trivial” since it is equivalent to trying all possible guesses for the key. We will next show that it is possible to reduce download cost significantly, by using *information sets* for Reed-Muller codes.

Definition 2. [12] Let $\mathcal{C}$ be an $[n, k]_q$ linear code and let $G \in \mathbb{F}_q^{k \times n}$ be a generator matrix for $\mathcal{C}$. Then, a set $\mathcal{I} \in \binom{[n]}{k}$ is an information set if $G_{\mathcal{I}} \in \mathbb{F}_q^{k \times k}$ is of rank k .

Intuitively, any codeword can be reconstructed in full by observing its entries on any information set. Furthermore, we have the following lemma that will be useful later. The proof

²Interestingly, recent works about privacy of quantized computation over the reals [7, 9, 17] employed a similar strategy of communicating a syndrome for saving upload costs, albeit in a very different problem setting.

follows from the definition of minimum distance of linear codes.

Lemma 1. *Let $\mathcal{C}$ be an $[n, k]_q$ linear code with minimum distance $d_{\min}$ and let $G \in \mathbb{F}_q^{k \times n}$ be a generator matrix for $\mathcal{C}$. Then, every subset of $[n]$ of size $n - d_{\min} + 1$ contains an information set for $\mathcal{C}$.*

We proceed with the definition of Reed-Muller codes [1].

Definition 3. [1, Def. 1] *For parameters d and m , the Reed-Muller code over $\mathbb{F}_q$ is defined as $RM_q(d, m) \triangleq \{\text{Eval}(f) \mid f \in \mathbb{F}_q[x], \deg(f) \leq d\}$, where $\deg(\cdot)$ denotes the total degree of a polynomial, and $\text{Eval}(f) = (f(\mathbf{z}))_{\mathbf{z} \in \mathbb{F}_q^m}$.*

The coordinates of $RM_q(d, m)$ are indexed using the q^m elements of $\mathbb{F}_q^m$, in an arbitrary order. In [12], an explicit description of information sets for Reed-Muller codes is provided. Specifically, by denoting $\mathbb{F}_q \triangleq \{\alpha_0, \dots, \alpha_{q-1}\}$, we have that $\mathcal{I}_{d,m} \triangleq \{(\alpha_{i_1}, \dots, \alpha_{i_m}) \mid \sum_{k=1}^m i_k \leq d, 0 \leq i_k \leq q-1\}$ is an information set for $RM_q(d, m)$. In addition, the next lemma follows directly from the definitions of information sets and Reed-Muller codes.

Lemma 2. *Let h be an m -variate polynomial over $\mathbb{F}_q$ of total degree d . If evaluations of h at an information set for $RM_q(d, m)$ are known, then it is possible to obtain evaluations of h at all other points in $\mathbb{F}_q^m$ by linearly combining the evaluations at the information set.*

We now present our scheme based on polynomial evaluation at information sets for Reed-Muller codes. The scheme assumes that the total degree of f is less than $m(q-1)$; that is, $d < m(q-1)$. Otherwise, $RM_q(d, m)$ is not well-defined.

The admin encodes $\tilde{\mathbf{x}}$ as defined in (1) to $\{\tilde{\mathbf{x}} - \mathbf{t}G \mid \mathbf{t} \in \mathcal{I}_{d,m}\}$ and distributes them to the workers. At a later point, the admin receives a multivariate polynomial f of total degree $d < m(q-1)$ from the user. Define a new multivariate polynomial $g : \mathbb{F}_q^m \rightarrow \mathbb{F}_q$ as

$$g(\mathbf{t}) \triangleq f(\tilde{\mathbf{x}} - \mathbf{t}G) \quad (2)$$

for all $\mathbf{t} \in \mathbb{F}_q^m$. It can be seen that the total degree of g is at most d , and that $g(\mathbf{k}) = f(\mathbf{x})$. The admin obtains

$$\mathcal{A} \triangleq (f(\tilde{\mathbf{x}} - \mathbf{t}G) \mid \mathbf{t} \in \mathcal{I}_{d,m}) \quad (3)$$

from the workers and sends it to the user. Since the user has now received evaluations of g at an information set for $RM_q(d, m)$, it follows from Lemma 2 that the user is able to compute $g(\mathbf{k}) = f(\mathbf{x})$ from $\mathcal{A}$ and the key $\mathbf{k}$.

It is straightforward to see that this scheme based on information set evaluations reduces download cost of the user with respect to the trivial scheme, provided that $d < m(q-1)$. To be precise, D now equals the size of the information set, $|\mathcal{I}_{d,m}|$, which is again equal to the dimension of the code $RM_q(d, m)$, written in the sequel as $\lambda(q, d, m)$. Note that to conclude the computation the user combines those $D = \lambda(q, d, m)$ elements linearly, which is normally much less costly than computing f locally since f is a polynomial on n variables.

Theorem 2. *Assuming $d < m(q-1)$, the scheme based on evaluations at information sets for Reed-Muller codes requires download cost $D = \lambda(q, d, m)$, where $\lambda(q, d, m)$ is the dimension of $RM_q(d, m)$.*

It is well known [1] that if $q = 2$, we have $\lambda(q = 2, d, m) = \sum_{i=0}^d \binom{m}{i}$, and [11] that if $d < q$, we have $\lambda(q, d < q, m) = \binom{m+d}{d}$. Hence, we have the following reduced download costs.

Corollary 1. *If $q = 2$ and $d < m$, then the scheme based on evaluation at information sets of Reed-Muller codes requires download cost $D = \sum_{i=0}^d \binom{m}{i}$. If $d < \min\{q, m(q-1)\}$, then the scheme based on evaluation at information sets of Reed-Muller codes requires download cost $D = \binom{m+d}{d}$.*

One example that demonstrates the benefit of using information sets is taking $\mathcal{C}^\perp$ as the $[n = 2^t - 1, k \geq n - \frac{r}{2}t, d_{\min} \geq r + 1]$ binary BCH code, where $t \geq 3$ is an integer and r is an even positive integer [15, Sec. 6.1]. The number m of parity symbols in $\mathcal{C}^\perp$, which equals the dimension of $\mathcal{C}$, is upper bounded by $\frac{r}{2} \log_2(n+1)$. This implies download cost $D \leq \sum_{i=0}^d \binom{\frac{r}{2} \log_2(n+1)}{i} \leq (1 + \frac{r}{2} \log_2(n+1))^d = \text{poly}(\log(n))$, assuming d is a constant and $r = O(\text{poly}(\log(n)))$. This download cost is a significant improvement over the trivial scheme with $D = 2^m > n$, at the same side information size of $m \leq \frac{r}{2} \log_2(n+1) = \text{poly}(\log(n))$. As a numeric example, suppose $n = 2^{14} - 1$, $r = 10$, and $d = 2$, which results in $m = 70$. The download cost is $D \leq 2486$ for our scheme, whereas $D = 2^{70}$ for the trivial scheme.

V. STRAGGLER-RESILIENT SYSTEMS

In this section, we seek to minimize the number of workers needed to compute the answer $\mathcal{A}$, in the presence of at most S stragglers. Clearly, $N = (S+1)D$ workers suffice to ensure that the admin can provide the answer in both schemes described in Section IV. This can be done by storing $(S+1)$ replicas of every $\tilde{\mathbf{x}} - \mathbf{t}G$, where $\mathbf{t} \in \mathbb{F}_q^m$ in the trivial scheme and $\mathbf{t} \in \mathcal{I}_{d,m}$ in the information set based scheme. Next, we show that it is possible to compute the answer with fewer workers than this repetition-based approach.

First, we use Lagrange Code Computing (LCC) [22], which requires large fields. Then, we establish the notion of *information super-sets*, based on information sets defined previously, and devise a scheme without any field size restriction.

A. LCC-based scheme

Recall that in the scheme described in Theorem 2, given a polynomial f , the admin employs the workers in order to compute $\mathcal{A} = (f(\tilde{\mathbf{x}} - \mathbf{t}G) \mid \mathbf{t} \in \mathcal{I}_{d,m})$. Let $\mathcal{I}_{d,m} = \{\mathbf{t}_1, \dots, \mathbf{t}_\lambda\}$, where $\lambda = \lambda(q, d, m)$. The LCC scheme can be applied as-is by the admin over the dataset $\{\tilde{\mathbf{x}} - \mathbf{t}_i G \mid i \in [\lambda]\}$. Namely, the admin constructs a univariate polynomial h of degree $\lambda - 1$ which evaluates to $\tilde{\mathbf{x}} - \mathbf{t}_1 G, \dots, \tilde{\mathbf{x}} - \mathbf{t}_\lambda G$ at designated $\mathbb{F}_q$ elements $\omega_1, \dots, \omega_\lambda \in \mathbb{F}_q$, evaluates h at N distinct evaluation points $\alpha_1, \dots, \alpha_N \in \mathbb{F}_q$ with $N = (\lambda - 1)d + S + 1$, and stores these evaluations in the workers, who apply f over them and return the results. The admin can then obtain $\mathcal{A}$

	Repetition	LCC	Lemma 4	[8]
Restriction	None	$q \geq N, d < m(q-1)$	$d < m(q-1)$	$q = 2, d < m$
N	$(S+1)\lambda$	$(\lambda-1)d + S + 1$	$q^m - d_{\min} + S + 1$	[8, Theorem 7]

TABLE I: Summary of various methods. Here $\lambda = \lambda(q, d, m)$ and $d_{\min}$ denotes the minimum distance of $RM_q(d, m)$.

while tolerating S stragglers. The LCC scheme, however, requires $N \leq q$.

Theorem 3. *Using the LCC scheme, an (n, q, r, d, S) -scheme exists with $N = (\lambda-1)d + S + 1$ workers, provided that $q \geq N$, where λ is the dimension of $RM_q(d, m)$. The download cost remains $D = \lambda$.*

We note that if $d < S$, then $(\lambda-1)d + S + 1 < (S+1)\lambda$. Next, we demonstrate another scheme based on the notion of *information super-sets* for Reed-Muller codes, defined shortly. The scheme has no field size restriction.

B. The information super-set scheme

We first define *information super-sets* for linear codes.

Definition 4. *Let $\mathcal{C}$ be an $[n, k, d_{\min}]_q$ linear code. Then, for any $0 < S < d_{\min}$, a multiset $\mathcal{T} \subseteq [n]$ is called an S -information super-set for $\mathcal{C}$ if every $(|\mathcal{T}| - S)$ -subset³ of $\mathcal{T}$ contains an information set for $\mathcal{C}$.*

Let g be the m -variate polynomial of degree at most d defined in (2). Suppose $\mathcal{T}$ is an S -information super-set for $RM_q(d, m)$, and the number of workers is equal to $|\mathcal{T}|$. The admin encodes $\tilde{\mathbf{x}}$ to $\{\tilde{\mathbf{x}} - \mathbf{t}G | \mathbf{t} \in \mathcal{T}\}$ and distributes them to the workers. From the definition of information super-sets, it follows that even in the presence of S stragglers, the admin receives the evaluations of g at an information set for $RM_q(d, m)$. It then follows from Lemma 2 that the admin can compute the answer $\mathcal{A}$ defined in (3). Let $L(q, d, m, S)$ denote the minimum size of an S -information super-set for $RM_q(d, m)$. We have the following result.

Theorem 4. *For parameters n, q, r, d, S , if there exists an $[n, m]_q$ linear code $\mathcal{C}$ whose dual code $\mathcal{C}^\perp$ has minimum distance at least $r+1$, and if $d < m(q-1)$, then there exists an (n, q, r, d, S) -scheme with $N = L(q, d, m, S)$ workers.*

Our information super-set based (n, q, r, d, S) -scheme for $d < m(q-1)$ can be described as follows. Let $\mathcal{T}$ be an S -information super-set for $RM_q(d, m)$ of size $L(q, d, m, S)$.

- The user encodes $\mathbf{x}$ to $\tilde{\mathbf{x}} \triangleq \mathbf{x} + \mathbf{k}G$ according to (1) using some random key $\mathbf{k} \sim K = \text{Unif}(\mathbb{F}_q^m)$ and sends $\tilde{\mathbf{x}}$ to the admin.
- The admin encodes $\tilde{\mathbf{x}}$ to $\hat{\mathbf{x}} = (\hat{\mathbf{x}}_1, \dots, \hat{\mathbf{x}}_N) = (\tilde{\mathbf{x}} - \mathbf{t}G | \mathbf{t} \in \mathcal{T})$ and distributes them to $N = L(q, d, m, S)$ workers.
- At a later point, the user sends f to the admin, who then sends it to the workers. The workers apply f on their data, and send the results back to the admin.

³We say that $\mathcal{S}$ is a k -subset of a multiset $\mathcal{T}$ if $\mathcal{S}$ is a multiset of size k , and the multiplicity of each element in $\mathcal{S}$ is at most its multiplicity in $\mathcal{T}$.

- The admin obtains $\{f(\tilde{\mathbf{x}} - \mathbf{t}G)\}_{\mathbf{t} \in \mathcal{I}}$, where $\mathcal{I} \subseteq \mathcal{T}$ is an information set of $RM_q(d, m)$, and sends $\mathcal{A} \triangleq (f(\tilde{\mathbf{x}} - \mathbf{t}G))_{\mathbf{t} \in \mathcal{I}, d, m}$ to the user.
- The user linearly combines the values in $\mathcal{A}$ to obtain $g(\mathbf{k}) = f(\mathbf{x})$.

Next, in order to understand the value of N in the above scheme, we construct explicit S -information super-sets. First, we present a lower- and upper-bound on the size of a minimum S -information super-set. The proofs of the following lemmas, which are given in the appendix, follow directly from the definition of information super-sets (Definition 4), and from Lemma 1.

Lemma 3. *$L(q, d, m, S) \geq \lambda(q, d, m) + S$, where $\lambda(q, d, m)$ is the dimension of $RM_q(d, m)$.*

Lemma 4. *Let $d_{\min}$ be the minimum distance of $RM_q(d, m)$. If $0 < S < d_{\min}$, then $L(q, d, m, S) \leq q^m - d_{\min} + S + 1$. In particular, if $q = 2$ then $L(q, d, m, S) \leq 2^m - 2^{m-d} + S + 1$.*

Corollary 2. *Let $d_{\min}$ be the minimum distance of $RM_q(d, m)$. If $d < m(q-1)$ and $0 < S < d_{\min}$, then the number of workers required in the information super-set scheme $N \leq q^m - d_{\min} + S + 1$. In particular, if $q = 2$ then $N \leq 2^m - 2^{m-d} + S + 1$.*

We also note that $L(q, d, m, S) \leq (S+1)\lambda(q, d, m)$, as repeating an information set $(S+1)$ times will give an S -information super-set.

Notions related to information super-sets have appeared previously in the literature. In [19], *computational locality* is studied for Reed-Muller codes. Additionally, some results regarding information super-set sizes are implied by existing works [11] about punctured Reed-Muller codes. We refer readers to the full version [8] for details.

Due to the importance of computations over $\mathbb{F}_2$, in the full version [8] we investigate the $q = 2$ case, and construct several explicit information super-sets which outperform the bound in Lemma 4. Our results on information super-sets for binary Reed-Muller codes can be summarized as follows. We come up with an explicit construction of a 1-information super-set for $RM_2(1, m)$, which yields $L(2, 1, m, 1) = m + 2$ if m is even and $L(2, 1, m, 1) \leq m + 3$ if m is odd. An explicit construction of a 2-information super-set for $RM_2(1, m)$ of size $2m + 1$ is also presented, and yet through a greedy computer search on a set-theoretic formulation we obtain smaller 2-information super-sets for $RM_2(1, m)$. Then, using the $(u, u+v)$ -construction of binary Reed-Muller codes [1], we come up with a construction of S -information super-sets for general binary Reed-Muller codes $RM_2(d, m)$ which outperforms Lemma 4 in most cases. In Table I, we give a summary of various methods used in this work in terms of their restrictions and the number of workers needed.

REFERENCES

- [1] E. Abbe, A. Shpilka, and M. Ye, "Reed–muller codes: Theory and algorithms," *IEEE Transactions on Information Theory*, vol. 67, no. 6, pp. 3251–3277, 2020.
- [2] K. Bhattad, K. R. Narayanan, *et al.*, "Weakly secure network coding," *NetCod, Apr*, vol. 104, pp. 8–20, 2005.
- [3] D. Catalano *et al.*, "Multiparty computation, an introduction," *Contemporary cryptology*, pp. 41–87, 2005.
- [4] A. Cohen, A. Cohen, M. Medard, and O. Gurewitz, "Secure multi-source multicast," *IEEE Transactions on Communications*, vol. 67, no. 1, pp. 708–723, 2018.
- [5] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd. New-York: Wiley, 2006.
- [6] P. Cuff and L. Yu, "Differential privacy as a mutual information constraint," in *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, 2016, pp. 43–54.
- [7] Z. Deng, V. Ramkumar, R. Bitar, and N. Raviv, "Private inference in quantized models," *arXiv preprint arXiv:2311.13686*, 2023.
- [8] Z. Deng, V. Ramkumar, and N. Raviv, "Perfect subset privacy in polynomial computation," *arXiv preprint arXiv:2405.05567*, 2024.
- [9] Z. Deng and N. Raviv, "Approximate private inference in quantized models," in *2023 IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2023, pp. 1597–1602.
- [10] M. Grassl, *Bounds on the minimum distance of linear codes and quantum codes*, Online available at <http://www.codetables.de>, Accessed on 2024-01-08, 2007.
- [11] V. Guruswami, L. Jin, and C. Xing, "Efficiently list-decodable punctured reed-muller codes," *IEEE Transactions on Information Theory*, vol. 63, no. 7, pp. 4317–4324, 2017.
- [12] J. D. Key, T. McDonough, and V. C. Mavron, "Information sets and partial permutation decoding for codes from finite geometries," *Finite Fields and Their Applications*, vol. 12, no. 2, pp. 232–247, 2006.
- [13] D. Kobayashi, H. Yamamoto, and T. Ogawa, "Secure multiplex coding attaining channel capacity in wiretap channels," *IEEE transactions on information theory*, vol. 59, no. 12, pp. 8131–8143, 2013.
- [14] K. Lee, M. Lam, R. Pedarsani, D. Papailiopoulos, and K. Ramchandran, "Speeding up distributed machine learning using codes," *IEEE Transactions on Information Theory*, vol. 64, no. 3, pp. 1514–1529, 2017.
- [15] S. Lin and D. J. Costello, *Error Control Coding: Fundamentals and Applications*. Pearson Prentice Hall, 2004.
- [16] B. Rassouli, F. E. Rosas, and D. Gündüz, "Data disclosure under perfect sample privacy," *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 2012–2025, 2019.
- [17] N. Raviv, R. Bitar, and E. Yaakobi, "Information theoretic private inference in quantized models," in *2022 IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2022, pp. 1641–1646.
- [18] N. Raviv and Z. Goldfeld, "Perfect subset privacy for data sharing and learning," in *2022 IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2022, pp. 1850–1855.
- [19] M. Rudow, K. Rashmi, and V. Guruswami, "A locality-based lens for coded computation," in *2021 IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2021, pp. 1070–1075.
- [20] S. Tarnopolsky and A. Cohen, "Coding-based hybrid post-quantum cryptosystem for non-uniform information," *arXiv preprint arXiv:2402.08407*, 2024.
- [21] C. Wang and N. Raviv, "Breaking blockchain's communication barrier with coded computation," *IEEE Journal on Selected Areas in Information Theory*, vol. 3, no. 2, pp. 405–421, 2022.
- [22] Q. Yu, S. Li, N. Raviv, S. M. M. Kalan, M. Soltanolkotabi, and S. A. Avestimehr, "Lagrange coded computing: Optimal design for resiliency, security, and privacy," in *The 22nd International Conference on Artificial Intelligence and Statistics*, PMLR, 2019, pp. 1215–1225.